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Summary

Machine learning engineer building large language model systems for technical domains, from model-level methods (prompt engineering, supervised fine-tuning, preference optimization) to production deployment (retrieval, inference serving, evaluation, reliability).

- Production LLM systems: owned retrieval performance and reliability for an enterprise LLM question-answering service in chip design and verification: NVIDIA NIM inference on Kubernetes, retrieval architecture redesign, HNSW tuning, and production debugging down to the C++ index layer.
- Structured generation and alignment: pipelines that turn unstructured domain text into schema-valid records; developer and implementer of TAB-PO, a token-level preference optimization method.
- Retrieval, agents, and evaluation: first author on evidence-to-taxonomy retrieval over a 17K-concept inventory; built the MCP-grounded auditing environment and its evaluation harness for an agentic financial benchmark, run against five agent frameworks and four backbones.

Domain experience spans semiconductor design automation (three engagements at Cadence Design Systems between 2021 and 2024), clinical NLP (Yale University, with clinicians at Yale–New Haven Hospital and Cleveland Clinic), financial AI (The Fin AI Community https://thefin.ai/index.html), and adversarial robustness in medical imaging (PhD work).

Education

University of Miami, Ph.D. in Computer Science, Coral Gables, FL — 2017 – 2023

- GPA: 3.85/4.0
- **Research Interest:** Robustness of deep neural networks, Medical Image/Signal Analysis

Institute of Software, Chinese Academy of Sciences, MS in Computer Science, Beijing, China — 2014 – 2017

- GPA: Top 10%
- **Research Interest:** Concurrent Software Testing

Northeastern University, BS in Information Security, Shenyang, China — 2010 – 2014

- GPA: Top 5%
- **Coursework:** C\C++, Java, Data Structure, Computer Architecture, Operating System, Computer Network, etc.

Programming Skills

- **Programming Language:** Python, C/C++, Java, C#, Ruby, R, SQL, Prolog, etc.
- **Framework:** Pytorch, Pandas, Scikit-learn, Numpy, Kubernetes, Docker, PyQT5, Java Swing, etc.

Experience

Postdoctoral Associate, Yale University – New Haven, CT — Jan 2025 – Present

- Built the extraction pipeline that turns free-text patient–provider portal messages into schema-valid records, each required to carry its semantic labels and an evidence span anchored in the source text.
- Designed field-level evaluation scoring label and evidence separately, which separates wrong-evidence from wrong-label failures; released as an annotated benchmark for hierarchical clinical extraction.
- Developed and implemented TAB-PO, a token-level preference optimization method that intervenes only at decoding positions below a confidence threshold.
- Ran the fine-tuning and preference-optimization on LLMs from 1.5B to 70B.

Software Engineer (T3), Machine Learning, Cadence Design Systems – San Jose, CA — Jul 2024 – Dec 2024

- Owned retrieval performance and reliability for the LLM question-answering service inside Cadence JedAI, the company's enterprise data and AI platform for chip design and verification, which served internal and external EDA engineers across tens of thousands of engineering documents and design data.
- Cut retrieval latency from ~10s to ~100ms by diagnosing a filter-pushdown mismatch in which metadata filters passed down from the application layer caused the vector store to bypass its HNSW index and linearly scan most of the corpus on every query; redesigned the filtering-to-index path so ordinary queries traverse the graph.
- Tuned HNSW parameters (M, ef_construction, ef_search) and embedding models as controlled experiments to trade recall against latency deliberately, evaluating every change on a validation set labeled by domain-expert application engineers.
- Diagnosed and resolved a production outage caused by a native-memory leak in the vector store's C++ HNSW layer: process RSS grew without bound while Python-side allocation tracking stayed flat, localizing the leak to an unreleased allocation on an exception path, which I patched in C++.
- Deployed NVIDIA NIM inference microservices on the team's Kubernetes cluster in direct collaboration with NVIDIA engineering.

Postdoctoral Associate, University of Florida – Gainesville, FL — Jan 2024 – Jun 2024

- Built a multi-modal deep learning system for clinical decision support in prostate tumor diagnosis, integrating medical imaging (micro-ultrasound) with structured clinical variables from patient records.
- Designed a CNN-based classification model for cervical vertebrae maturation (CVM) stage assessment from medical images, achieving 76.7% accuracy.
- Developed generative models (diffusion-based) for cross-modality medical image translation, enabling data augmentation and domain adaptation across imaging protocols.

Research Assistant, University of Miami – Coral Gables, FL — Sep 2019 – Dec 2023

- Developed robust deep learning methods for medical image analysis, including MR image segmentation, X-ray landmark detection, and blood cell object detection.
- Designed adaptive adversarial training algorithms with sample-specific margin adjustment to improve model robustness on medical imaging tasks while preserving classification and segmentation accuracy.
- Proposed a regularization-based framework to enhance CNN robustness for ECG signal classification under noisy conditions.
- Systematically evaluated robustness methods across diverse medical imaging modalities and architectures, demonstrating consistent generalization across clinical vision tasks.

Intern Software Engineer in Machine Learning, Cadence Design Systems, Inc. – San Jose, CA — May 2022 – Aug 2022

- Designed and implemented a machine learning system to accelerate worst-case measurement evaluation of integrated circuits (ICs), reducing discovery time by over 90% compared to traditional simulation-based methods.
- Built Gaussian Process regression models in Python/PyTorch, trained on simulation data sampled via Latin Hypercube Sampling.
- Applied Bayesian optimization to iteratively select simulation inputs most likely to produce worst-case measurements, dynamically updating the training dataset.
- Performed forward feature selection to remove noisy input features and reduce simulation outliers.

Intern Software Engineer in Machine Learning, Cadence Design Systems, Inc. – San Jose, CA — May 2021 – Aug 2021

- Designed a machine learning system to predict integrated circuit (IC) simulation outputs using Python and Scikit-learn.
- Built a data pipeline to preprocess historical simulation data into structured training sets.
- Implemented a Random Forest model to handle highly diverse simulation input ranges, achieving a mean

absolute error of 0.99 picoseconds.
- Developed an interactive UI with PyQt5 and Matplotlib to visualize feature impact on simulation outcomes.

Research Assistant, University of Miami and Northwestern University – Coral Gables, FL — Jan 2018 – Aug 2019
- Identified evolution patterns of successful academic groups through large-scale data mining.
- Parsed author and publication records (1991–2018) from the Microsoft Graph database using SQL, and constructed million-scale annual coauthor networks with Python and NetworkX.
- Designed a Monte Carlo-based clustering algorithm in Python and R to detect author groups within networks.
- Defined group evolution patterns (e.g., splitting, merging) across adjacent years.
- Analyzed the correlation between group evolution and academic success (e.g., citation counts) using statistical methods in Python and R.

Research Assistant, Institute of Software, Chinese Academy of Sciences – Beijing, China — Sep 2015 – May 2017
- Developed a multi-threaded C++ test case generator for validating concurrent data structures.
- Proposed and implemented three adaptive generation strategies, increasing concurrency error detection by up to 6% while reducing execution time by up to 10%.

Intern Software Engineer (C/C++), Hillstone Networks – Beijing, China — Jul 2015 – Nov 2015
- Migrated an Intrusion Prevention System (IPS) to new server hardware and updated the application stack using Ruby and Ruby on Rails.
- Implemented a packet sniffer in C/C++ to capture and process network traffic periodically for threat analysis.
- Automated backend SQLite operations with a tool chain using Python, SQL, and C/C++, enabling programmatic database updates in response to client requests.
- Implemented an HTTP URI filter in the firewall (C/C++) to detect and block abnormal or suspicious requests.

Intern Software Engineer (Java), Huaxin Education Technology Co., Ltd. – Shenyang, China — Jun 2013 – Aug 2013
- Developed a web application vulnerability scanner in Java/JSP with Oracle backend, capable of detecting common attacks such as SQL injection and cross-site scripting (XSS).
- Built a user interface in Java Swing to improve usability and streamline vulnerability reporting.

Selected Review Experience

- NeurIPS (2022 – 2025)
- ICML (2022, 2025, 2026)
- ICLR (2024 – 2026)
- AISTATS (2026)
- AAAI (2026)
- AI
- Neurocomputing

Teaching Experience

Teaching Assistant, CB&B 750 01 (SP25): Core Topics in Biomedical Informatics – Yale University — Spring 2025

Teaching Assistant, CSC546: Introduction to Machine Learning – University of Miami — Spring 2023

Teaching Assistant, CSC315: Introduction to Python for Scientists – University of Miami — Spring 2023

Teaching Assistant, CSC546: Introduction to Machine Learning – University of Miami — Fall 2022

Teaching Assistant, CSC315: Introduction to Python for Scientists – University of Miami — Spring 2020

Teaching Assistant, CSC315: Introduction to Python for Scientists – University of Miami — Fall 2019

Teaching Assistant, CSC546: Introduction to Machine Learning – University of Miami — Spring 2019

Teaching Assistant, CSC421: Computer Operating Systems – University of Miami — Fall 2018

Teaching Assistant, CSC421: Computer Operating Systems – University of Miami — Fall 2017

Publications

Factorized Hypothesis Search for Evidence-to-Taxonomy Retrieval — Submitted to ARR

L. Ma, E. F. Wei, X. Peng, Y. Wang, L. Qian, V. Gutiérrez-Basulto *https://arxiv.org/pdf/2608.06614*

EPPC-OASIS: Ontology-Aware Adaptation and Structured Inference Refinement for Electronic Patient-Provider Communication Mining in Secure Messages — Submitted to JBI

S. Fodeh, S. Ramachandran, E. Irankhah, M. Arif, A. Khan, G. Puthiaraju, **L. Ma**, S. Talakokkul, J. Alpert, S. Schellhorn *https://arxiv.org/pdf/2605.24172*

Herculean: An Agentic Benchmark for Financial Intelligence — Submitted to ARR

X. Peng, Z. Xie, Y. Cao, H. Li, L. Qian, Y. Wang, V. J. Zhang, H. He, X. Ai, **L. Ma**, et al. (60 authors) — Auditing workflow: task definition, environment, and evaluation design. *https://arxiv.org/pdf/2605.14355*

STaR-DRO: Stateful Tsallis Reweighting for Group-Robust Structured Prediction — Under Review

S. Fodeh, G. Puthiaraju, E. Irankhah, ***L. Ma***, S. Talakokkul, A. Khan, S. Ramachandran, J. Alpert, S. Schellhorn
https://arxiv.org/pdf/2604.09737

TAB-PO: Preference Optimization with a Token-Level Adaptive Barrier for Token-Critical Structured Generation — Submitted to Nature Machine Intelligence

S. Fodeh, ***L. Ma***, G. Puthiaraju, S. Talakokkul, A. Khan, A. Hagaman, S. Lowe, A. Roundtree
https://arxiv.org/pdf/2603.00025

PVminerLLM: Structured Extraction of Patient Voice from Patient-Generated Text using Large Language Models — 2026

S. Fodeh, ***L. Ma***, G. Puthiaraju, S. Talakokkul, A. Khan, A. Hagaman, S. Lowe, A. Roundtree
NPJ Digital Public Health, Accepted

EPPCMinerBen: A Novel Benchmark for Evaluating Large Language Models on Electronic Patient-Provider Communication via the Patient Portal — 2026

S. Fodeh, Y. Wang, ***L. Ma***, S. Talakokkul, J. Alpert, S. Schellhorn
Artificial Intelligence in Medicine

PVminer: A Domain-Specific Tool to Detect the Patient Voice in Patient Generated Data — Under Review

S. Fodeh, ***L Ma***, Y Wang, S Talakokkul, G. Puthiaraju, A. Khan, A. Hagaman, S. Lowe, A. Roundtree

Scientific Report, Under Review

Attention-based Shape-Deformation Networks for Artifact-Free Geometry Reconstruction of Lumbar Spine from MR Images 2025

L Qian, J Chen, ***L Ma***, T Urakov, W Gu, L Liang

IEEE Transactions on Medical Imaging

Intrathread Method Orders Based Adaptive Testing of Concurrent Objects 2025

Y Dai, L Xie, P Wu, S Cui, ***L Ma***

Science of Computer Programming, 103362

A novel attention-based network for geometry reconstruction with error estimation from medical images 2025

L Qian, J Chen, ***L Ma***, T Urakov, W Gu, L Liang

Medical Imaging 2025: Image Processing 13406, 169–176

Symtc: A symbiotic transformer-cnn net for instance segmentation of lumbar spine MRI 2024

J Chen, L Qian, ***L Ma***, T Urakov, W Gu, L Liang

Computers in Biology and Medicine, 179, 108795

Adaptive Testing of Concurrent Objects 2024

Y Dai, P Wu, S Cui, ***L Ma***

Theoretical Aspects of Software Engineering (TASE 2024), Guiyang, China

Evaluation of Deep Neural Network Models for Instance Segmentation of Lumbar Spine MRI 2024

J Chen, L Qian, ***L Ma***, T Urakov, W Gu, L Liang

bioRxiv, 2024.04.02.587810

A sequential geometry reconstruction based deep learning approach to improve accuracy and consistence of lumbar spine MRI image segmentation 2024

L Qian, J Chen, ***L Ma***, T Urakov, L Liang

Medical Imaging 2024: Image Processing 12926, 753–760

A general approach to improve adversarial robustness of DNNs for medical image segmentation and detection 2024

L Ma, J Chen, L Qian, L Liang

Medical Imaging 2024: Image Processing 12926, 266–274

Towards Improving the Adversarial Robustness of Deep Neural Networks 2023

L Ma

University of Miami Dissertation

Improving Adversarial Robustness of Deep Neural Networks via Adaptive Margin Evolution 2023

L Ma, L Liang

Neurocomputing, 551, 126524

Increasing-Margin Adversarial (IMA) Training to Improve Adversarial Robustness of Neural Networks 2023

L Ma, L Liang

Computer Methods and Programs in Biomedicine, 240, 107687

Towards lifting the trade-off between accuracy and adversarial robustness of deep neural networks for medical image classification and segmentation 2023
L Ma, L Liang
Medical Imaging 2023: Image Processing 12464, 429–436

Adaptive Adversarial Training to Improve Adversarial Robustness of DNNs for Medical Image Segmentation and Detection 2023
L Ma, L Liang
arXiv:2206.01736

A Regularization Method to Improve Adversarial Robustness of Neural Networks for ECG Signal Classification 2022
L Ma, L Liang
Computers in Biology and Medicine, 144, 105345

Enhance CNN Robustness Against Noises for Classification of 12-Lead ECG with Variable Length 2020
L Ma, L Liang
19th IEEE International Conference on Machine Learning and Applications (ICMLA 2020)

An Algorithm for Out-Of-Distribution Attack to Neural Network Encoder 2020
L Liang, ***L Ma***, L Qian, J Chen
arXiv:2009.08016

Improve robustness of DNN for ECG signal classification: a noise-to-signal ratio perspective 2020
L Ma, L Liang
ICLR 2020 Workshop on AI for Affordable Health

Diversity driven adaptive test generation for concurrent data structures 2018
L Ma, P Wu, T.Y. Chen
Information and Software Technology, 103, 162–173